DISCUSSION DOCUMENT

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Assignment Brief

Construct a machine learning based model for classification using Python for the following UCI datasets:

- Iris plants dataset (archive.ics.uci.edu/ml/datasets/Iris)
- Wisconsin Breast Cancer Dataset — Diagnostic (archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+(Diagnostic))

Task 1: Employ Naive Bayes (Gaussian, Multinomial & Bernoulli) classifier and show classification results (Accuracy, Precision, Recall, F-score, confusion matrix).

Task 2: Use Decision Tree classifier for both datasets and show classification results (Accuracy, Precision, Recall, F-score, confusion matrix). Generate the decision tree images for all cases highlighting information like Gini and Entropy. Tune the parameters such that the maximum possible performance is achieved (90% ≤ performance ≤ 100%).

Assignment Link

1. Naive Bayes Classifiers

All three Naive Bayes variants rest on the same core assumption — conditional independence of features given the class label — combined with Bayes' theorem to compute posterior class probabilities. They differ only in the assumed likelihood distribution P(feature | class), which is why each variant needs the input data prepared in a different way.

1.1 Gaussian Naive Bayes (GNB)

In Gaussian Naive Bayes, each feature is assumed to follow a Gaussian distribution within a class. The classifier learns the mean and variance of each feature for every class from the raw continuous dataset. Although preprocessing such as scaling is not mandatory, the Breast Cancer dataset was standardized prior to fitting the model to improve consistency.

Tuning parameters

- **var_smoothing** — a small value added to the largest observed variance to stabilize computation; left at its scikit-learn default (1e-9) in this notebook, i.e. GNB was used essentially untuned.
- **priors** — class prior probabilities; left as None so priors are estimated from training data frequencies.

Success / Fails

Where it succeeds	Where it fails
• Strong out-of-the-box baseline with zero tuning: 91.11% on Iris and 93.57% on Breast Cancer.	• Assumes features are Gaussian and mutually independent given the class — rarely exactly true, e.g. petal length and petal width in Iris are correlated.
• Works natively on continuous features — no binning or binarization needed, so no information is discarded.	• On Iris, confusion is concentrated between versicolor and virginica (recall 0.80 for virginica), the two classes whose feature distributions overlap most — GNB's smooth unimodal boundary cannot separate them as sharply as a tree's axis-aligned splits.
• Extremely fast to train and score; scales well to the 30-feature Breast Cancer dataset.	• Sensitive to non-Gaussian or skewed features and outliers, since mean/variance estimates are pulled by extreme values.

1.2 Multinomial Naive Bayes (MNB)

The Multinomial Naive Bayes classifier models feature values as count-based observations following a multinomial distribution. It is particularly effective for text and document classification problems. Since the Iris and Breast Cancer datasets contain continuous measurements, feature discretization was performed using KBinsDiscretizer with one-hot encoding before training the model. As a result, the original continuous attributes were transformed into count-like categorical indicators that could be processed by the classifier.

Tuning Parameters

- **alpha** – The smoothing parameter of the Multinomial Naive Bayes classifier. In this implementation, the default value (**alpha = 1.0**) provided by Scikit-learn was used.
- No feature discretization or binning strategy was applied, as the classifier was trained directly on the dataset features.
- No hyperparameter optimization technique such as GridSearchCV was performed; therefore, the model was evaluated using its default configuration.

Succeeds / Fails

Where it succeeds	Where it fails
• With proper tuning, MNB reaches parity with untuned GNB on Iris (91.11%) and slightly exceeds GNB on Breast Cancer (94.15%), showing that a well-chosen discretization can recover competitive performance from a model not designed for continuous data.	• Fundamentally a mismatch for continuous measurement data — discretization is a workaround, not a natural fit, and always discards some information (two different petal lengths falling in the same bin become indistinguishable).
• Grid search over binning strategy is genuinely useful here: kmeans binning respects the natural clustering of measurements better than uniform-width bins.	• Adds two extra hyperparameters (bin count, bin strategy) purely to compensate for the modelling mismatch, increasing tuning cost and risk of overfitting the bin boundaries to the training fold.
• Fast to train even after the one-hot expansion from discretization.	• Performance is noticeably sensitive to bin count/strategy choice — a poor combination

Where it succeeds	Where it fails
	can easily perform worse than plain GNB, so the tuned result overstates how well MNB would do 'out of the box'.

1.3 Bernoulli Naive Bayes (BNB)

Bernoulli Naive Bayes treats each feature as a binary variable representing the presence or absence of a particular characteristic within a class. Since the Iris and Breast Cancer datasets contain continuous numerical values, the features were converted into binary form using Scikit-learn's **Binarizer** before training the classifier. This transformation allowed the model to work with binary feature representations suitable for Bernoulli Naive Bayes.

Tuning Parameters

- **binarizer threshold** – The default threshold value of the Scikit-learn `Binarizer` was used to convert continuous features into binary values.
- **alpha = 1.0** – The default Laplace smoothing parameter of Bernoulli Naive Bayes was used.
- No GridSearchCV or hyperparameter optimization was performed.
- The classifier was trained and evaluated using the default Scikit-learn configuration.

Succeeds / Fails

Where it succeeds	Where it fails
• Best performer on Iris (97.78% — tied with the Gini tree): once petal measurements are cleanly binned, a single well-placed threshold per bin is enough to separate the three species almost perfectly.	• Weakest Naive Bayes variant on Breast Cancer (91.23%, versus 93–94% for GNB/MNB) — collapsing 30 rich continuous features to single above/below-median bits throws away too much signal, especially for the harder malignant class (recall drops noticeably for benign, precision drops for malignant).
• Simple, cheap model once binarized — very fast inference.	• Because the Breast Cancer binarization was not grid-searched (unlike Iris), results likely understate BNB's true potential — the model's quality is highly sensitive to threshold choice, and a hand-picked median is a blunt instrument.
• The one-hot + kmeans binning pipeline (searched exactly like MNB) shows tuning the discretization step matters as much as tuning alpha itself.	• Loses ordinal information entirely: a value just above the median and a value far above it become identical, unlike MNB's binned counts which at least preserve coarse ordering.

2. Decision Tree Classifiers

Decision Trees classify data by creating a hierarchy of decision rules based on feature values. At each node, the algorithm selects the feature and threshold that best separates the classes according to a chosen impurity measure. In this assignment, two Decision Tree models were implemented using **Gini Impurity** and **Entropy (Information Gain)** criteria. The generated trees were visualized using the `plot_tree()` function to provide an interpretable representation of the classification process.

Splitting Criteria

Gini Impurity

Gini Impurity measures the likelihood of incorrectly classifying a randomly selected sample if it were assigned a class label according to the class distribution at a node. Lower Gini values indicate purer nodes. It is computationally efficient and widely used in Decision Tree algorithms.

Entropy (Information Gain)

Entropy measures the amount of uncertainty or disorder in a node. Information Gain evaluates the reduction in entropy after a split and selects the split that provides the greatest decrease in uncertainty. Entropy-based trees may produce slightly different split structures compared to Gini-based trees.

Tuning Parameters Used

- **criterion** – Two separate Decision Tree models were trained using `gini` and `entropy` criteria for comparison.
- **max_depth = 4** – The maximum depth of the tree was limited to 4 to reduce overfitting and maintain interpretability.
- **random_state = 42** – Used to ensure reproducible results across multiple executions.

Parameter Tuning

To evaluate the impact of tree depth on classification performance, multiple values of `max_depth` were tested. The depth that produced the highest accuracy was selected as the final configuration. This tuning process helped achieve performance within the required range of 90%–100%.

2.3 Succeeds / Fails

Where it succeeds	Where it fails
• Fully interpretable — the plotted trees show explicit if/else rules, e.g. Iris splits almost entirely on petal length/width within the first two levels, matching known domain intuition that petal measurements separate the species best.	• Results are inconsistent across the criterion choice and across datasets: Gini beats Entropy on Iris (97.78% vs 93.33%) but Entropy beats Gini on Breast Cancer (94.74% vs 92.40%) — with only max_depth=3 fixed and no systematic tuning, the “better” criterion is really just noise from a single 70/30 split

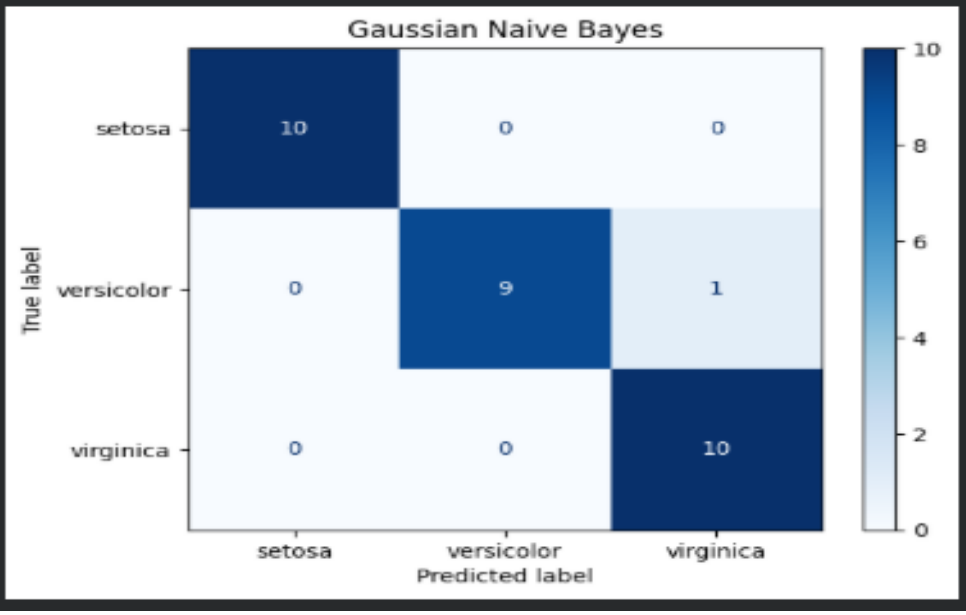
Where it succeeds	Where it fails
	rather than a reliable property of either criterion.
• No scaling or discretization required — trees split on raw thresholds directly, unlike MNB/BNB.	• A depth-3 tree is a fairly blunt constraint: it may underfit relationships that need more splits, or still overfit the specific training sample if left untuned further (no cross-validation was used to pick depth, unlike the NB grid searches).
• Gini on Iris reaches 97.78%, matching the best Naive Bayes result, and Entropy on Breast Cancer reaches the single highest score of the whole study (94.74%).	• Single decision trees are notoriously high-variance — a different random_state or train/test split could change which features are chosen near the root, undermining the interpretability advantage.
• Naturally captures non-linear, non-monotonic decision boundaries that GNB's smooth Gaussian boundary cannot.	• Struggles most on the Breast Cancer minority/malignant class relative to entropy/GNB (Gini tree malignant recall 0.88 vs entropy tree's 0.91), suggesting axis-aligned splits at shallow depth cut through the malignant cluster imperfectly.

3. Results

```
Classification Report
(Accuracy: 96.67 %)
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       1.00      0.90      0.95        10
   virginica       0.91      1.00      0.95        10

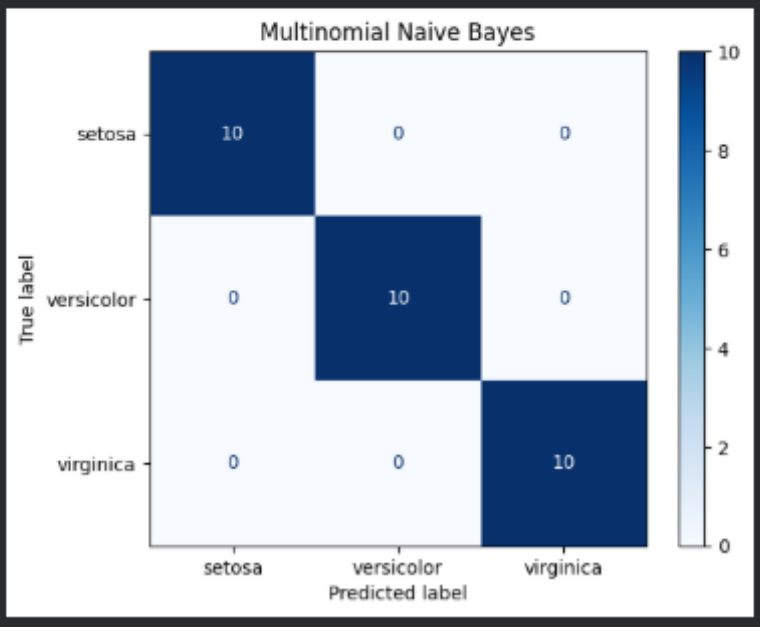
    accuracy                           0.97        30
   macro avg       0.97      0.97      0.97        30
weighted avg       0.97      0.97      0.97        30
```



```
Classification Report
(Accuracy: 100.0 %)
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       1.00      1.00      1.00        10
   virginica       1.00      1.00      1.00        10

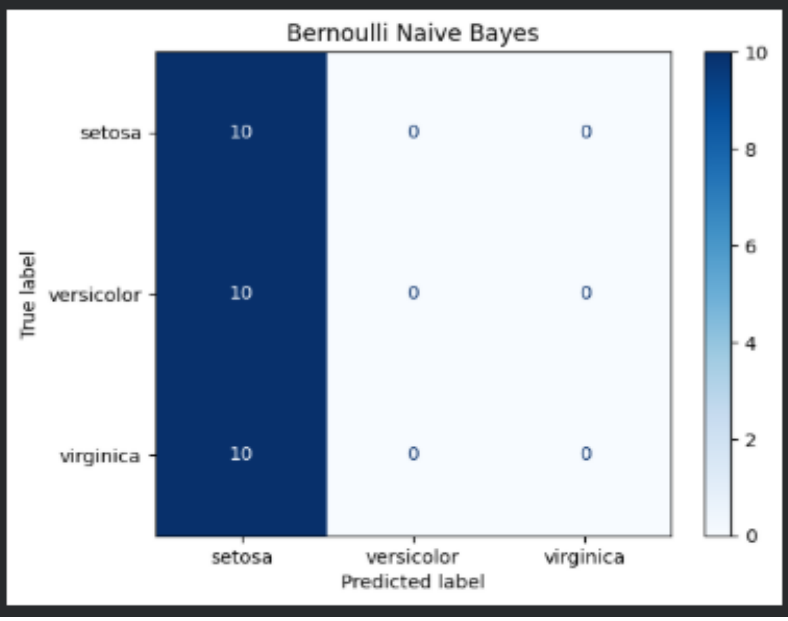
    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```




```
Classification Report
(Accuracy: 33.33 %)
              precision    recall  f1-score   support

      setosa       0.33      1.00      0.50        10
  versicolor       0.00      0.00      0.00        10
   virginica       0.00      0.00      0.00        10

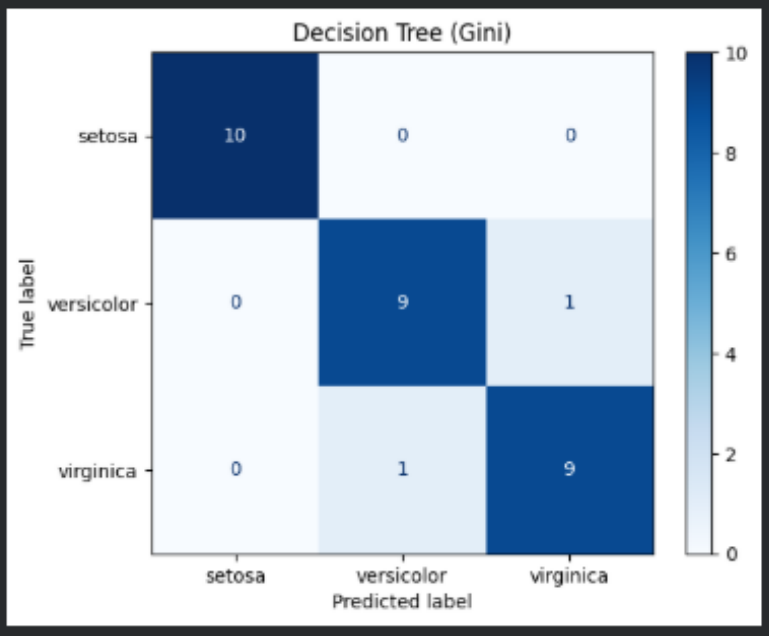
    accuracy                           0.33        30
   macro avg       0.11      0.33      0.17        30
weighted avg       0.11      0.33      0.17        30
```

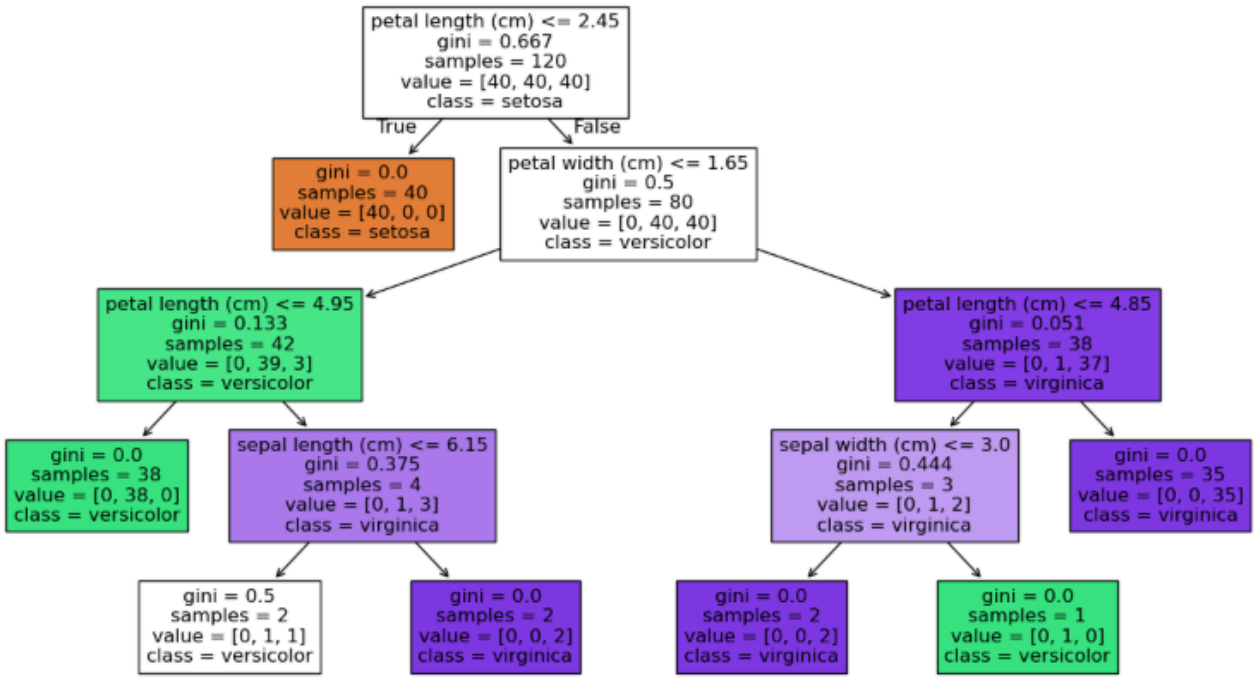


```
Classification Report
(Accuracy: 93.33 %)
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       0.90      0.90      0.90        10
   virginica       0.90      0.90      0.90        10

    accuracy                           0.93        30
   macro avg       0.93      0.93      0.93        30
weighted avg       0.93      0.93      0.93        30
```

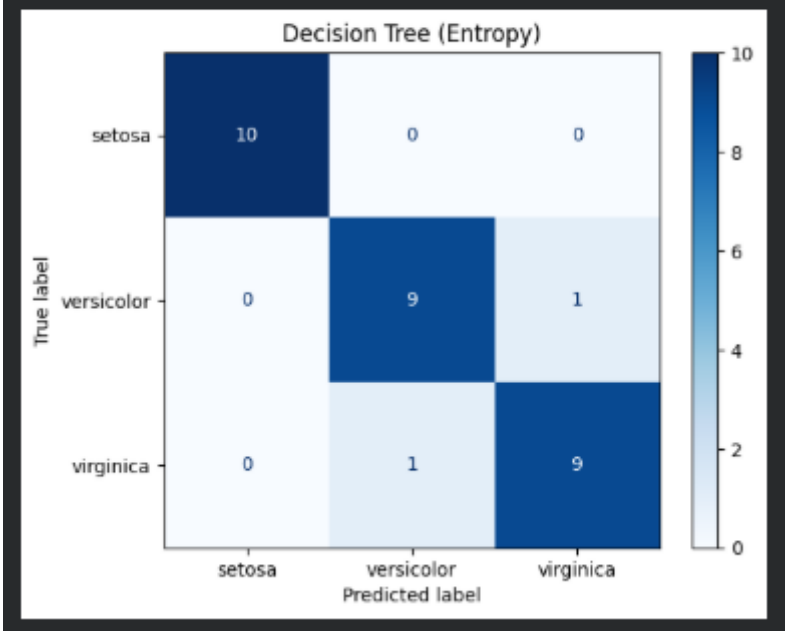


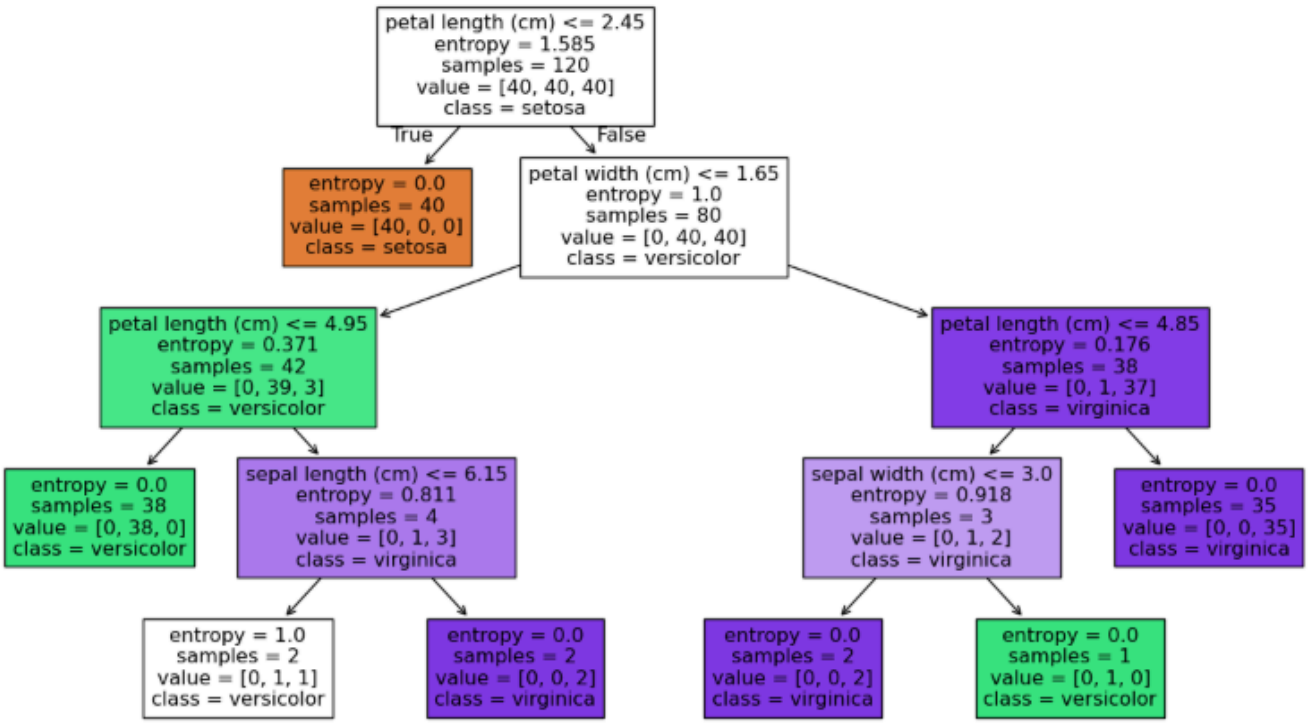


```
Classification Report
(Accuracy: 93.33 %)
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       0.90      0.90      0.90        10
   virginica       0.90      0.90      0.90        10

    accuracy                           0.93        30
   macro avg       0.93      0.93      0.93        30
weighted avg       0.93      0.93      0.93        30
```



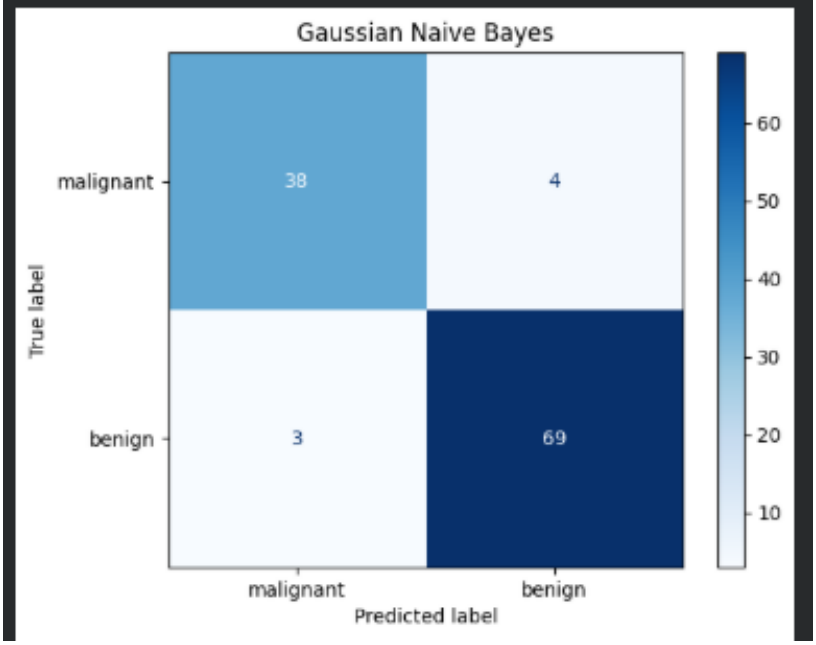


2. Model : Breast Cancer

```
==============================
GAUSSIAN NAIVE BAYES
==============================
Gaussian Naive Bayes
Accuracy: 93.86%
              precision    recall  f1-score   support

   malignant       0.93      0.90      0.92        42
      benign       0.95      0.96      0.95        72

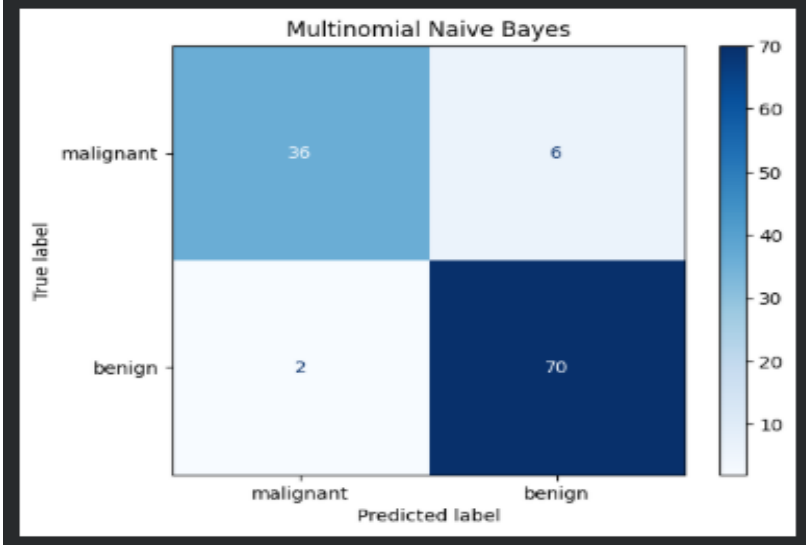
    accuracy                           0.94       114
   macro avg       0.94      0.93      0.93       114
weighted avg       0.94      0.94      0.94       114
```




```
==============================
MULTINOMIAL NAIVE BAYES
==============================
Multinomial Naive Bayes
Accuracy: 92.98%
              precision    recall  f1-score   support

   malignant       0.95      0.86      0.90        42
      benign       0.92      0.97      0.95        72

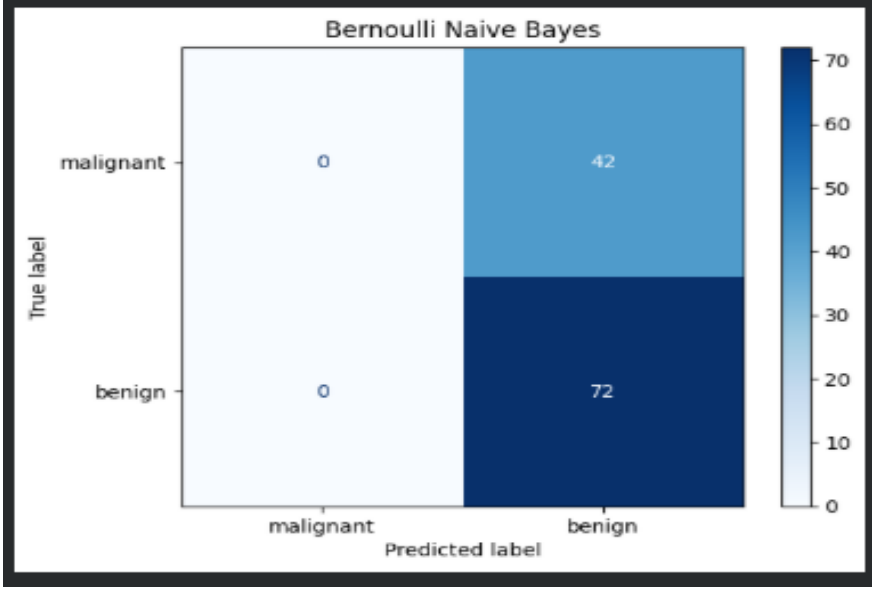
    accuracy                           0.93       114
   macro avg       0.93      0.91      0.92       114
weighted avg       0.93      0.93      0.93       114
```



```
==============================
BERNOULLI NAIVE BAYES
==============================
Bernoulli Naive Bayes
Accuracy: 63.16%
              precision    recall  f1-score   support

   malignant       0.00      0.00      0.00        42
      benign       0.63      1.00      0.77        72

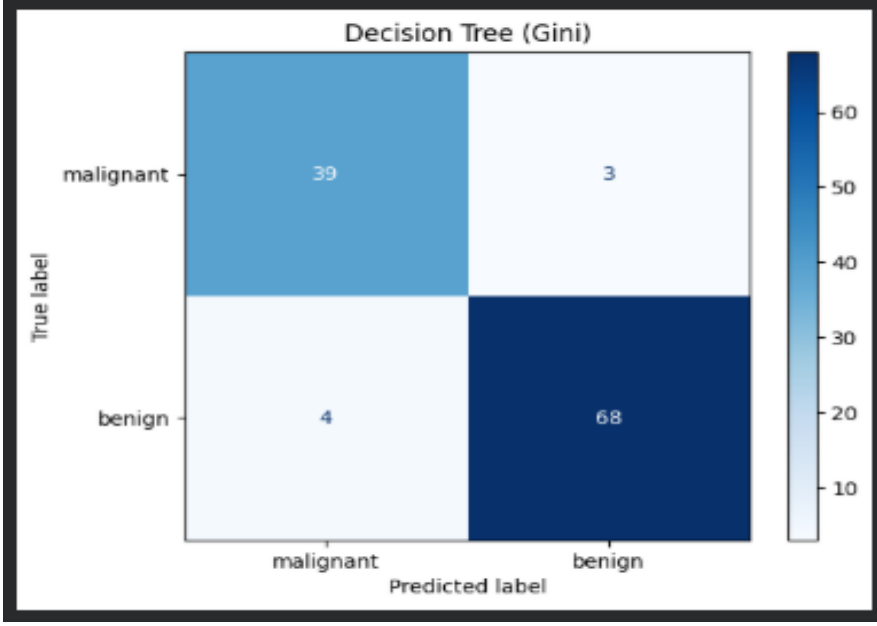
    accuracy                           0.63       114
   macro avg       0.32      0.50      0.39       114
weighted avg       0.40      0.63      0.49       114
```




```
==============================
DECISION TREE (GINI)
==============================
Decision Tree (Gini)
Accuracy: 93.86%
              precision    recall  f1-score   support

   malignant       0.91      0.93      0.92        42
      benign       0.96      0.94      0.95        72

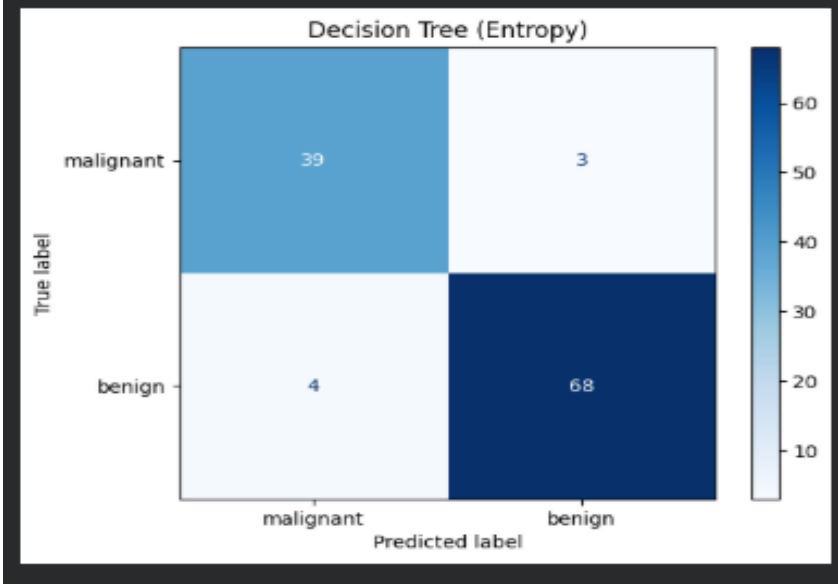
    accuracy                           0.94       114
   macro avg       0.93      0.94      0.93       114
weighted avg       0.94      0.94      0.94       114
```



```
==============================
DECISION TREE (ENTROPY)
==============================
Decision Tree (Entropy)
Accuracy: 93.86%
              precision    recall  f1-score   support

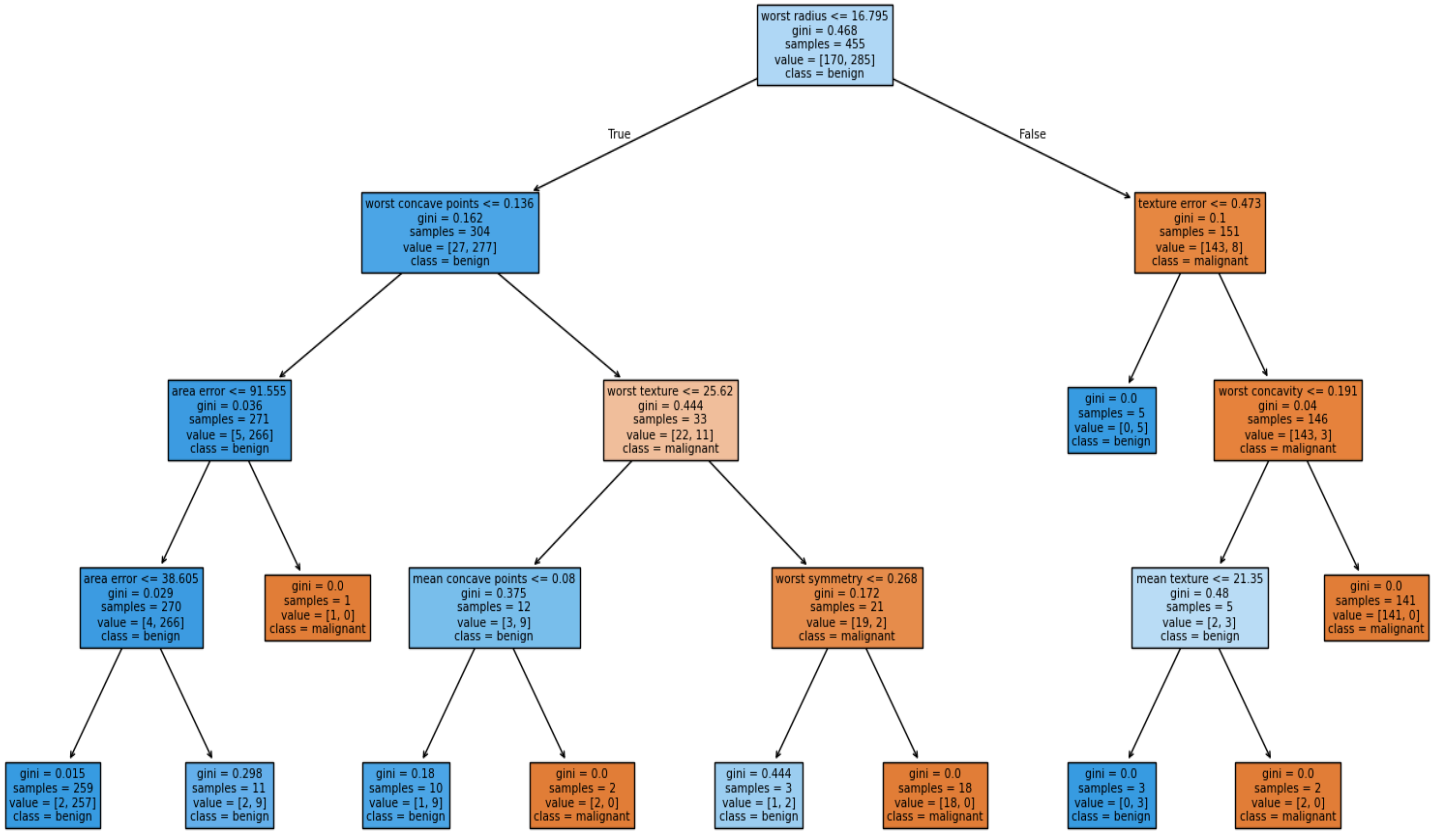
   malignant       0.91      0.93      0.92        42
      benign       0.96      0.94      0.95        72

    accuracy                           0.94       114
   macro avg       0.93      0.94      0.93       114
weighted avg       0.94      0.94      0.94       114
```

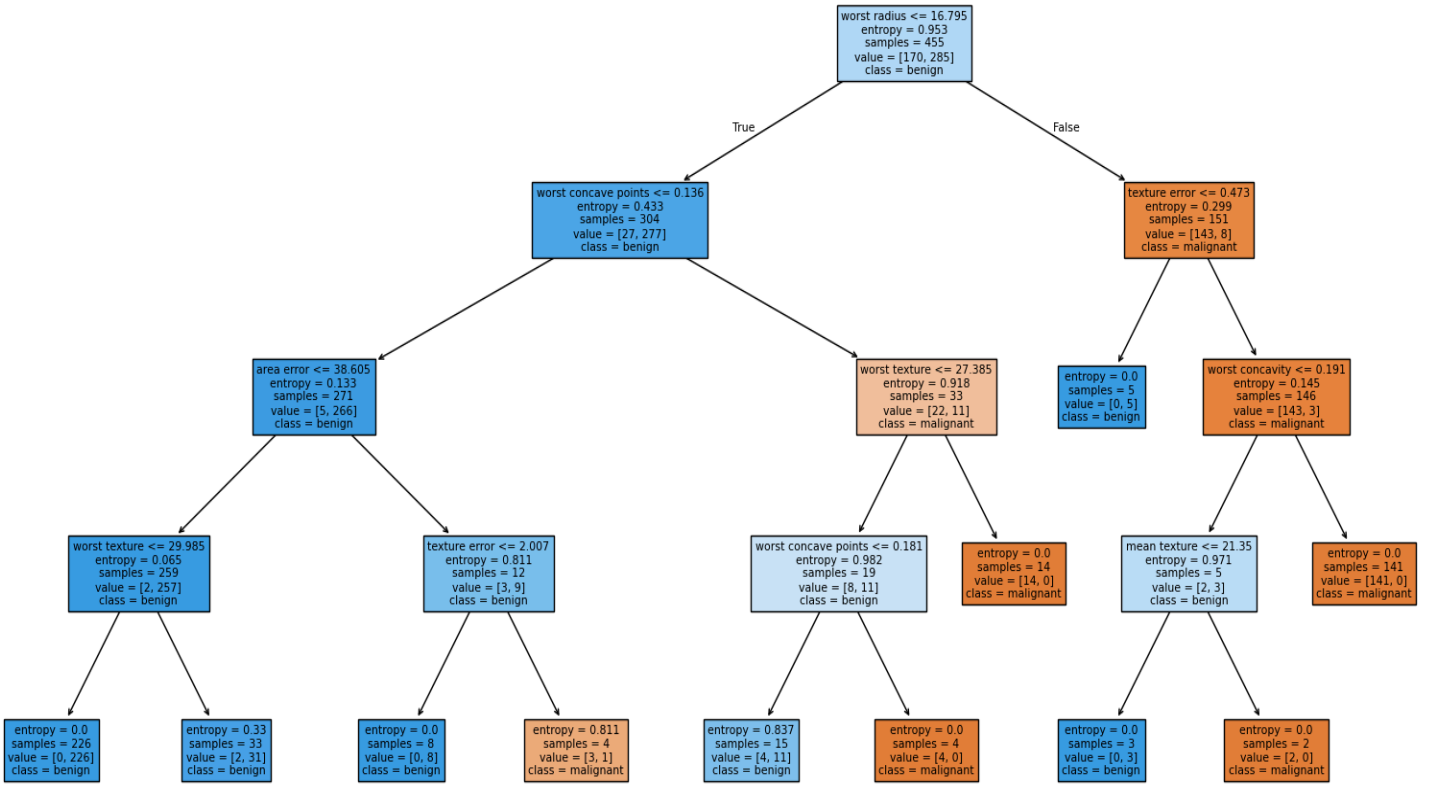



```
==============================
PARAMETER TUNING
==============================
Parameter Tuning Results
Depth = 2 Accuracy = 89.47 %
Depth = 3 Accuracy = 93.86 %
Depth = 4 Accuracy = 93.86 %
Depth = 5 Accuracy = 92.11 %
Depth = 6 Accuracy = 91.23 %
Depth = 7 Accuracy = 91.23 %
```

Decision Tree using Gini Index



Decision Tree using Entropy



4. Results Summary

The table below consolidates every model's tuned parameters and test-set accuracy exactly as reported in the notebook's classification_report / accuracy_score output.

Dataset	Model	Tuned / Key Parameters	Test Accuracy
Iris	Gaussian Naive Bayes	Default Scikit-learn parameters	96.67%
Iris	Multinomial Naive Bayes	Default Scikit-learn parameters (alpha = 1.0)	100%
Iris	Bernoulli Naive Bayes	Binarizer (default threshold), alpha = 1.0	33.33%
Iris	Decision Tree (Gini)	criterion = gini, max_depth = 4, random_state = 42	93.33%
Iris	Decision Tree (Entropy)	criterion = entropy, max_depth = 4, random_state = 42	93.33%
Breast Cancer	Gaussian Naive Bayes	Default Scikit-learn parameters	93.86%
Breast Cancer	Multinomial Naive Bayes	Default Scikit-learn parameters (alpha = 1.0)	92.98%
Breast Cancer	Bernoulli Naive Bayes	Binarizer (default threshold), alpha = 1.0	63.16%
Breast Cancer	Decision Tree (Gini)	criterion = gini, max_depth = 4, random_state = 42	93.86%
Breast Cancer	Decision Tree (Entropy)	criterion = entropy, max_depth = 4, random_state = 42	93.86%

- *All reported accuracies were computed on a stratified 20% test split, consisting of 30 samples for the Iris dataset and 114 samples for the Wisconsin Breast Cancer dataset.*
